

Operation Doomsday — Internal Walkthrough

Internal Use Only

Confidential — Do Not Distribute.

This walkthrough contains the full solutions, exploitation sequences, and flag texts.

Use only for verification and challenge authoring.

Flag 1 — SATID Spoofing

Issue: SAT_ID=0x00 incorrectly routes to D00M02.

Command:

```
doomgs uplink-raw d00d00010100020001012511
```

Expected: Mode change on D00M02.

Flag: FLAG{SATID_SPOOF_PWN}

Flag 2 — Secure Crosslink Crypto Bug

Underlying flaw: XOR hash-based crypto with nonce leaks.

Command:

```
doomgs xlink-send 2 17 2001
```

Expected: K00KIES01 decrypts and becomes compromised.

Flag: FLAG{CROSSLINK_CRYPTOPWN}

Flag 3 — Firmware Signature Weakening

K00KIES02 checks only first 8 hex chars of firmware signature.

Upload malicious firmware starting with 'DOOMFW!!'.

Apply with matching prefix.

Flag: FLAG{FIRMWARE_BACKDOOR_ON_ORBIT}

Flag 4 — RF Sniffer

K00KIES■01 has debug_rx_all=True.

Intercept blocked crosslinks.

Flag: FLAG{SNIFFED_THROUGH_THE_MASK}

Flag 5 — RF Attenuation

Power <20% or RAAN misalignment breaks RF delivery.

Manipulate with:

doomgs uplink-nudge-power 2 -90

Flag: FLAG{ATTENUATION_AND_ELATION}

Flag 6 — Thermal Overflow

Repeated temp offsets trigger safe mode.

Commands:

doomgs uplink-temp-offset 1 +80

doomgs uplink-temp-offset 1 +80

Flag: FLAG{TOO_HOT_TO_HANDLE}

Flag 7 — Replay Attack

Reuse sequence numbers with --seq override.

Replay executes despite detection.

Flag: FLAG{REPLAY_THAT_SLAPS}

Flag 8 — Orbit Drift Manipulation

CMD 0x04 interprets RAAN delta as unsigned.

doomgs uplink-drift-raan 1 240

Flag: FLAG{RAAN_TILTED_THE_SCALE}

Final Flag — Constellation Domination

All satellites compromised=True.

System triggers final beacon.

Flag: FLAG{MF_DOOM_CONTROLS_ORBIT}